

Ryan Chiou

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EDUCATION

University of Illinois Urbana-Champaign

Expected Dec 2026

Bachelor of Science in Mechanical Engineering

GPA: 3.6/4.0

- **Coursework:** Robotics, Product Design & Additive Manufacturing, Design Optimization, Mechanical Design II

EXPERIENCE

IBM

May 2026 – Aug 2026

Hardware Developer Intern | Data & Production Automation

New York, NY

- Designed enclosures and mounts in SolidWorks for an iPhone-based end-of-line inspection rig for IBM Z mainframes
- Cut projected inspection system cost by 88% by replacing a \$250K industrial robot with a \$30K iPhone rig
- Deployed a Python/PostgreSQL/React dashboard tracking 8-figure material flow to fully automate a 4-hr process

Tesla

May 2025 – Dec 2025

Supplier Industrialization Engineer Intern | Battery & Energy Plastics

Fremont, CA

- Drove \$350K annual savings on injection-molded parts via DFM/DFA, tooling optimization, process development, and material sourcing across Tesla's V4 Supercharger, Powerwall 3, Semi battery, Megapack, and Solar products
- Owned the DFM iteration loop between design release and supplier tooling for 100+ plastic parts, driving rib redesigns, wall thickness tuning, and shrinkage compensation that eliminated warpage, sink, and short-shot risk
- Directed Moldflow analyses of process and boundary conditions to validate melt flow and dimensional conformity
- Root-caused 50+ part defects and drove design/process corrective actions, cutting defect escalations by 95%

Illini Electric Motorsports

Aug 2023 – Jul 2025

Engineering Project Lead | Formula SAE Electric

Urbana-Champaign, IL

- Spearheaded redesign of inverter discharge PCB and energy monitoring sensor enclosures, reducing the mass-to-volume ratio by 12% while increasing component capacity and enabling sub-30-second service accessibility
- Engineered an accumulator cart with an integrated pawl-and-slot deadman brake mechanism; validated the frame's structural integrity against buckling failure under a 300 lbf load with a FoS of 5 using Ansys FEA
- Optimized undertray venturi profile in Creo using STAR-CCM+ CFD; increased downforce and reduced mass by 9%
- Fabricated 10+ carbon fiber parts via prepreg layups with fixed-volume fractions on CNC-milled polyurethane molds

PROJECTS

3D Scanning Robotic Arm | Robotics Mechanisms & Controls

- Designed a 6-DOF arm equipped with a Helios2+ Narrow ToF camera for real-time 3D scanning using point clouds
- Optimized arm segment structures to target 500 mm+ reach and 1 mm positional repeatability on a \$2K budget
- Implemented forward/inverse kinematics in C++ with real-time microcontroller joint control for teleoperation
- Developed an Arena SDK/OpenCV/Open3D point cloud pipeline that aligns arm multi-pose scans using forward kinematics and hand-eye calibration, fusing them via ICP into an accurate, unified 3D model

Single-Input Planetary Walker | Robotics Mechanisms

- Designed a 10-bar linkage leg with path synthesis, tracing a foot path with a 50 mm flat stance and 32 mm lift
- Performed kinematic and dynamic force analysis in Python to size leg link geometry and drivetrain gear ratios
- Packaged robot in Fusion 360 within a 300×180×180 mm envelope and prototyped iterations using FDM 3D printing
- Integrated a Geneva-driven manipulator that sequences payload actuation off a single-DOF motor input

Deep Sea ROV | Underwater Robotics

- Led the design of a 0.5 kg Delrin ROV frame housing an IP68 electronics enclosure with 6-DOF vectored thrusters
- Integrated Arduino + Raspberry Pi control stack for PWM thruster control and 1080p onboard video over Ethernet
- 3D printed resin camera enclosures, validated by a vacuum hold and 5 m depth submersion at 4× operating duration

TECHNICAL SKILLS

Design & Analysis: SolidWorks, Creo Parametric, Fusion 360, nTopology, Moldflow, Ansys FEA, DFM/DFA, GD&T

Fabrication: FDM & Resin 3D Printing, Injection Molding, CNC/Manual Machining, Laser Cutting, Carbon Fiber Layups

Software & Controls: Python, MATLAB, C/C++, ROS 2, Arduino, Raspberry Pi, SQL, React, JavaScript